



Strong yet tough cross-linked oligosiloxane elastomer with impact-resistant and antifouling properties

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ABSTRACT

Polyurea with strong hydrogen bond interactions has excellent mechanical strength, toughness, and strain-rate-dependent behavior. However, the too fast polymerization and lack of functions limit its applications. In this study, a high-strength polyurea elastomer with impact resistance and antifouling function was prepared with secondary amine–oligosiloxane nanoclusters and a diisocyanate-terminated polytetrahydrofuran (PTMG)-based prepolymer, where the nanoclusters provide mechanical strength and antifouling ability while the urea segments give elasticity and toughness. Due to the lower reaction activity of secondary amine groups, the polymerization rate decreases, so that the operation time can be extended to more than 60 min without gelation. The polyurea elastomers exhibit excellent tensile strength (7.7–28.2 MPa) and strain rate sensitivity (10.8–16.6 MPa). The dynamic mechanical tests demonstrate that it has good energy dissipation capacity ($\tan \delta > 0.3$ at 25–30 °C). Bioassays show that the polyurea elastomers exhibit excellent protein resistance and can effectively inhibit the adhesion of microorganisms (bacteria *S. aureus*, *E. coli*, and *Pseudomonas* sp., as well as diatom *N. incerta*). The polyurea elastomers exhibit good versatility and can be combined with Kevlar fabric through a simple dip-coating process to form a composite with puncture resistance force as high as 108.8 N, 10 times higher than that of the one-layer Kevlar fabric. The prepared polyurea elastomers are expected to find applications in flexible wear protection, marine antifouling, and biomedical engineering.

1. Introduction

Polyurea is synthesized through the polyaddition of isocyanate components and amino compounds [1–3]. It exhibits excellent mechanical strength, toughness, and strain-rate-dependent behavior due to strong hydrogen bond interactions [4–9]. As a result, polyurea has received considerable attention in the fields of protective coatings, adhesives, biomedical engineering, flexible electronics, ballistic proof materials and wear-resistant materials [10–17]. However, the polyaddition between isocyanate and primary amino is very fast (mostly 5–10 s) limiting its large-scale production and application [18,19]. To decrease the polymerization rate, polyurea is usually synthesized at a low temperature and many strong polar solvents are dropwise added [20,21]. In industrial application, specialized spraying equipment has been manufactured to solve this problem [22]. However, the equipment is inconvenient to transport and operate. Particularly, the polyurea properties cannot be regulated in spraying.

Using secondary amines such as polyaspartic ester (PAE) synthesized via primary amines and maleic acid ester is effective to reduce the

polymerization rate due to its steric hindrance and electron induced effect [23]. By adjusting the type of PAE [24], the gel time of polyurea can be tuned from several minutes to more than two hours. Yet, PAE-based polyurea has lower hydrogen bond density compared with polyurea synthesized by primary amines, which declines the mechanical properties of the polyurea. To improve the mechanical strength, high content of hard segments is used in PAE-based polyurea, but such polyurea is brittle with low elongation at break. Alternatively, nanofillers such as nano titanium dioxide, and carbon nanotubes are incorporated into PAE-based polyurea to improve the mechanical strength, but it may cause sedimentation during the curing process [25,26]. Note that the aforementioned polyurea without functional groups is not capable of anti-biofouling. It cannot meet some practical application in the fields of flexible wear protection, marine antifouling, and biomedical engineering, where the antifouling functions are critical besides high mechanical strength [27]. So far, it is a great challenge to develop polyurea with regulatable operation time, high mechanical properties, impact resistance and antifouling ability [28].

In this study, a strong yet tough cross-linked polyurea elastomer with

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impact-resistant and antifouling properties has been prepared by combining sol–gel chemistry and isocyanate–secondary amine addition reaction. First, two types of secondary amine–oligosiloxane nanoclusters (one for the matrix and the other for antifouling) were synthesized via hydrolysis and condensation of N-phenylaminomethyltriethoxysilane and silane-terminated amphiphilic telomers. Subsequently, a diisocyanate-terminated PTMG-based prepolymer was used to link the nanoclusters at room temperature. The elastomers have a high cross-linking density because of the abundant secondary amine on the oligosiloxane nanoclusters, which can provide excellent mechanical properties. The presence of nanoclusters and hydrogen bonds can endow the polyurea elastomers with outstanding impact resistance. Meanwhile, the polyurea elastomers contains amphiphilic telomers, making them possible to resist biofouling [29–31]. The relations between the structures and mechanical properties of the polyurea elastomers including strain-rate-dependent behavior, wettability and antifouling property have been investigated. To verify their versatility, the composite based on Kevlar fiber and polyurea elastomers was prepared. The puncture resistance and antifouling performance of composite were also examined. Our aim is to provide a simple but valuable strategy to prepare high-performance polyurea elastomer with impact resistance, antifouling ability, and reasonable operation time.

2. Experimental section

2.1. Materials

Dodecafluoroheptyl methacrylate (DFMA, $M_n = 400$ g/mol), poly(ethylene glycol) methyl ether methacrylate (PEGMA, $M_n = 475$ g/mol), isophorone diisocyanate (IPDI), dibutyltin dilaurate (DBTDL), (3-mercaptopropyl) trimethoxysilane (KH590), N-phenylaminomethyltriethoxysilane, polytetramethylene glycol (PTMG, $M_n = 1000$ g/mol), 4,4'-diaminodiphenylmethane, *n*-hexane, azobisisobutyronitrile (AIBN), and γ -aminopropyl triethoxysilane (KH550) were purchased from Aladdin. AIBN was purified by recrystallizing twice in methanol before use. Artificial seawater (ASW) was prepared according to ASTM D1141-98 (2003). The ammonia solution ($\text{NH}_3\cdot\text{H}_2\text{O}$, 25–28 wt %) and tetrahydrofuran (THF) were from Sinopharm. THF was distilled by CaH_2 before use. Fibrinogen ($M_w = 340$ kDa) was from Merck Chemicals. Physiological phosphate buffered saline (PBS, pH = 7.4) was prepared according to published article [32].

2.2. Synthesis of amphiphilic Telomers

As shown in Scheme S1, KH590 (3.92 g), PEGMA (19.00 g), DFMA (16.00 g), AIBN (0.08 g) and THF (40 mL) were introduced in a Schlenk tube and degassed through three freeze – evacuate – thaw cycles. The molar ratio of PEGMA to DFMA is 1/1. Then, nitrogen was introduced into the tube to maintain an anhydrous and oxygen-free environment. After reaction at 70 °C for 24 h, the solution was precipitated in *n*-hexane. After removing *n*-hexane with a vacuum oven, a colorless transparent liquid was obtained. The successful synthesis of the telomers was confirmed by ^1H NMR (Fig. S1). According to the integral area of 5.44 ppm, 4.08 ppm, 2.36 ppm, the molecular weight of the telomer is about 1900 g/mol.

2.3. Synthesis of N-phenylaminomethylsilsesquioxane Nanoclusters

N-phenylaminomethyltriethoxysilane (21.00 g), deionized water (2.08 g), ammonia solution (0.60 g), and THF (30 mL) were added into a three-neck flask and stirred at 60 °C for 8 h. After the reaction, THF was removed by rotary evaporation and the obtained nanoclusters are named NS. The successful synthesis of NS was confirmed by ^{29}Si NMR (Fig. S2).

2.4. Synthesis of functionalized Nanoclusters

N-phenylaminomethyltriethoxysilane (21.00 g), deionized water (2.08 g), ammonia solution (0.60 g), and THF (30 mL) were added to a three-neck flask and stirred at 60 °C for 4 h. Subsequently, amphiphilic telomers (7.00 g), H_2O (0.12 g), and ammonia solution (0.10 g) were added to the mixture. After stirring for another 4 h, THF was removed, and the obtained nanoclusters are designated as NA. The successful synthesis of NA was also confirmed by ^{29}Si NMR (Fig. S3).

2.5. Synthesis of Diisocyanate-Terminated PTMG-Based Prepolymer

PTMG (20.00 g) was heated at 110 °C in a three-neck flask under vacuum and stirred for 2 h to remove water. After cooling to 80 °C, IPDI (8.89 g) and DBTDL (0.06 g) were introduced into the flask under nitrogen atmosphere. The prepolymer was yielded after a 3-h reaction at 80 °C [33].

2.6. Preparation of polyurea Elastomers

As depicted in Scheme 1, the polyurea elastomers were prepared by using two types of secondary amine–oligosiloxane nanoclusters (NS and NA) and a diisocyanate-terminated PTMG-based prepolymer. Briefly, NS and NA were dissolved in THF and added to the prepolymer. After stirring for 30 min at 25 °C, the mixture was poured into a polytetrafluoroethylene plate and cured at 25 °C for 3 d. The compositions of the polyurea elastomers are listed in Table 1. For all samples, the molar ratio of NH/NCO exceeded 1. Note that we use the mass percentage of each substance in the Table 1. This is because the structure of nanoclusters (NS and NA) is stereoscopic, we cannot obtain the actual molecular weight, thus it is not possible to show the molar ratio between the two components.

2.7. Preparation of Kevlar/Polyurea Composite

Kevlar fabric was washed with a sodium dodecyl sulfate solution (1 wt%) and rinsed with distilled water several times [34]. After drying at 70 °C for 12 h, the polyurea/Kevlar composite was prepared through a dip-coating process. Kevlar fabric was cut into square samples ($1 \times 1 \text{ cm}^2$ and $10 \times 10 \text{ cm}^2$) and soaked in the polyurea solution (50 wt%) three times. Afterward, the samples were cured at 25 °C for 3 d.

2.8. Characterization

Nuclear magnetic resonance spectroscopy (^1H NMR and ^{29}Si NMR) analysis was performed on a spectrometer (Bruker AV600 NMR) using CDCl_3 as the solvent and tetramethylsilane as the internal standard. The fourier transform infrared spectra (FT-IR) was recorded on a VERTEX 70 spectrometer (Bruker) with 64 scans at a resolution of 4 cm^{-1} .

2.9. Wide-Angle X-ray diffraction (XRD) Test

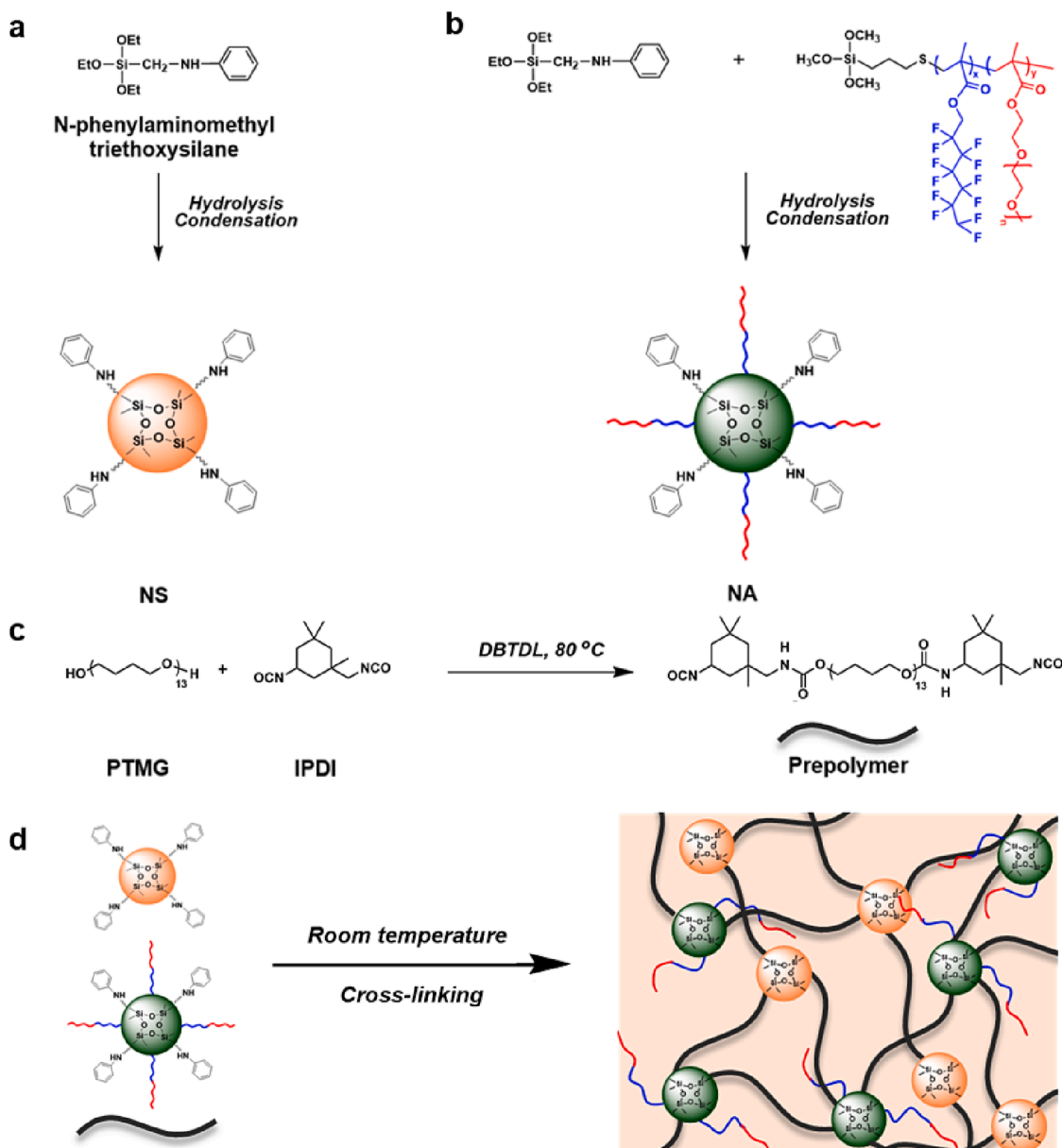
The thin-film XRD experiments with samples were carried out on a X'pert Powder (PANalytical) with Cu target, scanning from (2θ) 2–90° with a scanning speed of 12°/min.

2.10. Atomic force microscopy (AFM) Test

NS and NA were added in THF (1 mg/L) and spin-coated on the silicon wafer. The test was performed in a non-contact mode using an XE-100 AFM (Park Systems, Korea) in air at 25 °C.

2.11. Tensile Test

The mechanical properties of the polyurea elastomers were evaluated through tensile tests using an Instron 5966 instrument with a 10-kN



Scheme 1. Schematic diagram of the synthesis of the polyurea elastomers.

Table 1
Compositions of NSx–NAy.

Sample	NS (wt%)	NA (wt%)	Prepolymer (wt%)
NS6–NA0	60	0	40
NS4–NA2	40	20	40
NS3–NA3	30	30	40
NS2–NA4	20	40	40
NS0–NA6	0	60	40

load cell at 25 °C. The elastomers were rectangular shaped with a 2-mm width, 0.2-mm thickness and 35-mm length. The stretching rate of 2–500 mm/min was selected. Cyclic tensile test was conducted by stretching the sample to 100% strain, and then unloading it to original position with a strain rate of 2–500 mm/min. The sample was successively loaded and unloaded within 5 cycles.

2.12. Dynamic mechanical Test

The dynamic mechanical tests were conducted on a dynamic mechanical analyzer (DMA) (KSV NIMA). The viscoelastic property was measured under nitrogen atmosphere with a heating rate of 10 °C/min from –80 to 150 °C. 1 Hz and 50 Hz were chosen as the test frequencies. The measurements were performed until the polyurea elastomers became too soft to be tested.

2.13. Falling ball impact Test

A self-constructed device was used to investigate the impact-resistant performance of the polyurea elastomers. The elastomers were cut into a square shape (5 × 5 cm²) with a thickness of 2 mm and placed on the sensor. A steel ball (17 g) was released at a height of 1 m (from the elastomer to the steel ball). The impact force was recorded by a computer, and each sample was tested three times to obtain an average value. The experimental group in which the falling ball directly hit the sensor (without elastomer on it) was selected as the control group.

2.14. Thermogravimetric analysis (TGA)

A thermogravimetric analyzer (NETZSCH, TG 209F1) was used to performed thermogravimetric analysis. Samples (5–8 mg) were placed in a clean crucible for TGA. The test temperature range was 30–900 °C, and the gas atmosphere was nitrogen. The heating rate was 10 °C/min. For comparison, polyurea without nanoclusters was prepared using diisocyanate-terminated PTMG-based prepolymer and 4,4'-diaminodiphenylmethane [35].

2.15. X-ray photoelectron spectroscopy (XPS)

XPS analysis was performed using an Axis Ultra DLD (Kratos Analytical) with a monochromate Al K α ($h\nu = 1486.6$ eV) source at takeoff angles of 90°. C (1 s) at 284.6 eV was used as the reference for the binding energy.

2.16. Water contact angle (WCA) measurement

The static WCA was measured on a theta optical tensiometer (Attension, Biolin Scientific) by dropping a droplet of liquid (3 μ L) on the surface of sample at 25 °C. The average value was obtained by measuring five different regions.

2.17. Quartz crystal microbalance with dissipation (QCM-D)

The protein resistance ability of the polyurea elastomers was investigated by QCM-D from Q-sense AB (Sweden). Polyurea films were prepared by spin-casting with polyurea solution in THF (5.0 mg/mL) on an AT-cut quartz crystal and cured at 25 °C for 24 h. Fibrinogen solution (1.0 mg/mL) was delivered to the film surface at a flow rate of 150 μ L/min using ASW as a reference. Briefly, the frequency change (Δf) is related to the protein adsorption, whereas the dissipation change (ΔD) is correlated to the viscoelastic properties of the adsorbed protein layer. More details of the QCM-D measurement can be found in previous studies [36–38].

2.18. Antibacterial Assay

Gram-positive bacteria (*S. aureus*), gram-negative bacteria (*E. coli*), and marine bacteria (*Pseudomonas* sp. NOV776) were used to evaluate antibacterial performance. The polyurea films were prepared on a silicon wafer (1×1 cm²) by spin-casting with a 24-h curing process at 25 °C and a blank silicon wafer was chosen as the control sample. Then, the samples were immersed in bacterial suspension (10⁷ CFU/mL) of ASW or PBS for 4 h at 30 °C in a 24-well microplate and washed with ASW (*Pseudomonas* sp.) or PBS (*S. aureus* and *E. coli*). Subsequently, the adhered bacteria were dyed by 1.5- μ L LIVE/DEAD BacLight Bacterial Viability Kit for 0.5 h and observed using a fluorescence microscope (Scope A1, Zeiss).

S. aureus and *E. coli* were also selected for fabric antibacterial tests. The composite and control sample (untreated Kevlar fabric) were immersed in bacterial suspension of PBS (10⁶ CFU/mL) and incubated for 4 h at 30 °C. Then, the sample was washed with deionized water and fixed for 1 h using a 3% glutaraldehyde solution of PBS. The sample was washed twice with PBS and once with deionized water, and dehydrated with alcohol of various concentrations (50%, 75%, 90%, and 100%) for 15 min, respectively. After drying overnight in a desiccator, the samples coated with platinum (at 30 mA for 1 min) were observed using a scanning electron microscope (SEM) under a magnification of $\times 2000$.

2.19. Anti-Diatom Test

The diatom *Navicula incerta* (*N. incerta*) obtained from Xiamen University was cultivated in an F/2 culture medium under a 12:12 h light–dark cycle at 23 °C in a light incubator. The polyurea films were

also prepared on a silicon wafer (1×1 cm²) by spin-casting with a 24-h curing process at 25 °C and a blank silicon wafer was chosen as the control sample. The samples were placed in a 24-well microplate and immersed in diatom suspension (a chlorophyll value of 0.25 μ g/L). After soaking for 24 h, ASW was used to wash the sample, and the adhered diatoms were observed using a microscope (Scope A1, Zeiss).

2.20. Puncture resistance measurement of Kevlar/Polyurea Composite

The puncture resistance measurement of the Kevlar/polyurea composite was evaluated using an Instron5966 with a 10-kN load cell at 25 °C. The samples were fixed using a self-constructed fixture. A sharp needle was used to pierce perpendicular to the composite at a piercing speed of 50 mm/min.

3. Results and discussion

Scheme 1 illustrates the preparation of the oligosiloxane nanoclusters (NS and NA), diisocyanate-terminated PTMG-based prepolymer and polyurea elastomers. For convenience, the polyurea elastomer was termed NS_x-NA_y, where x and y refer to the mass ratio of NS and NA to the total mass of all substance, respectively. FT-IR and XRD tests were carried out to examine the molecular structure of nanoclusters. As shown in Fig. S4, 960 cm^{−1} attributed to the Si-OCH₂CH₃ bond disappeared, and 1045 and 1150 cm^{−1} attributed to the Si-O-Si bond showed a significant increase in intensity, which are usually observed in ladder-structured polysilsesquioxanes [39]. In the XRD patterns of NS6-NA0 and NS0-NA6, there is a set of two distinct diffraction peaks at $2\theta = 5^\circ$ and 20° (Fig. S5), further indicating the ladder-structured [40]. In addition, Fig. S6 shows the AFM and SEM images of NS and NA, the morphology of both nanoclusters are irregular spheres and the size of the two nanoclusters is about 20–80 nm. To verify that the secondary amine-oligosiloxane nanoclusters can reduce the reaction rate, primary amine-oligosiloxane nanoclusters were synthesized according to previous literature [41]. Under the same conditions, the prepolymer solution will not flow when the primary amine-oligosiloxane nanoclusters were added (3 ~ 5 s). But the prepolymer solution containing secondary amine-oligosiloxane nanoclusters can be stirred for 1 h at 25 °C.

Fig. 1a depicts the stress–strain curves of NS_x-NA_y. NS6-NA0 has the highest tensile strength of 28.2 MPa. As NA content increases, the tensile strength decreases to 7.7 MPa. This is because the high cross-linking density via nanoclusters decreases as NS is gradually replaced by NA, so the mechanical properties of the NS_x-NA_y decline [42]. Considering NS3-NA3 has appropriate mechanical properties and antifouling properties at the same time, NS3-NA3 was chosen as the typical sample for the follow-up experiments. Fig. 1b and 1c show the cyclic tensile test of NS3-NA3. After 5 cycles of tension, NS3-NA3 still holds intact and exhibits high mechanical strength. After a single stretch session, NS3-NA3 almost returns to its original state with a ten-minute recovery, indicating that NS_x-NA_y has not only high tensile strength but also excellent resilience and fatigue resistance. Moreover, the NS3-NA3 elastomer (~5-mm width, ~0.2-mm thickness) can withstand a 10-kg dumbbell for more than 10 min without breaking (Fig. 1d). This is due to the abundant hydrogen bonds in the nanoclusters.

Fig. 2 shows the results of dynamic mechanical measurements. The peak of tan δ shows an obvious upward trend as NA content increases (Fig. 2a). This is because the cross-linking density decreases as NS is gradually replaced by NA, which endows the polymer segments with improved mobility, thereby increasing the energy dissipation capacity of elastomers [43]. NS3-NA3 was also chosen as a typical sample to test on the DMA at 1 Hz and 50 Hz. Its storage modulus increases with the test frequency at 0–90 °C (Fig. 2b). As the frequency increases from 1 Hz to 50 Hz, the peak of tan δ rises significantly and shifts to higher temperatures. Generally, a material with tan δ greater than 0.3 is considered to have good energy absorption ability and buffering effects [44]. Here, the

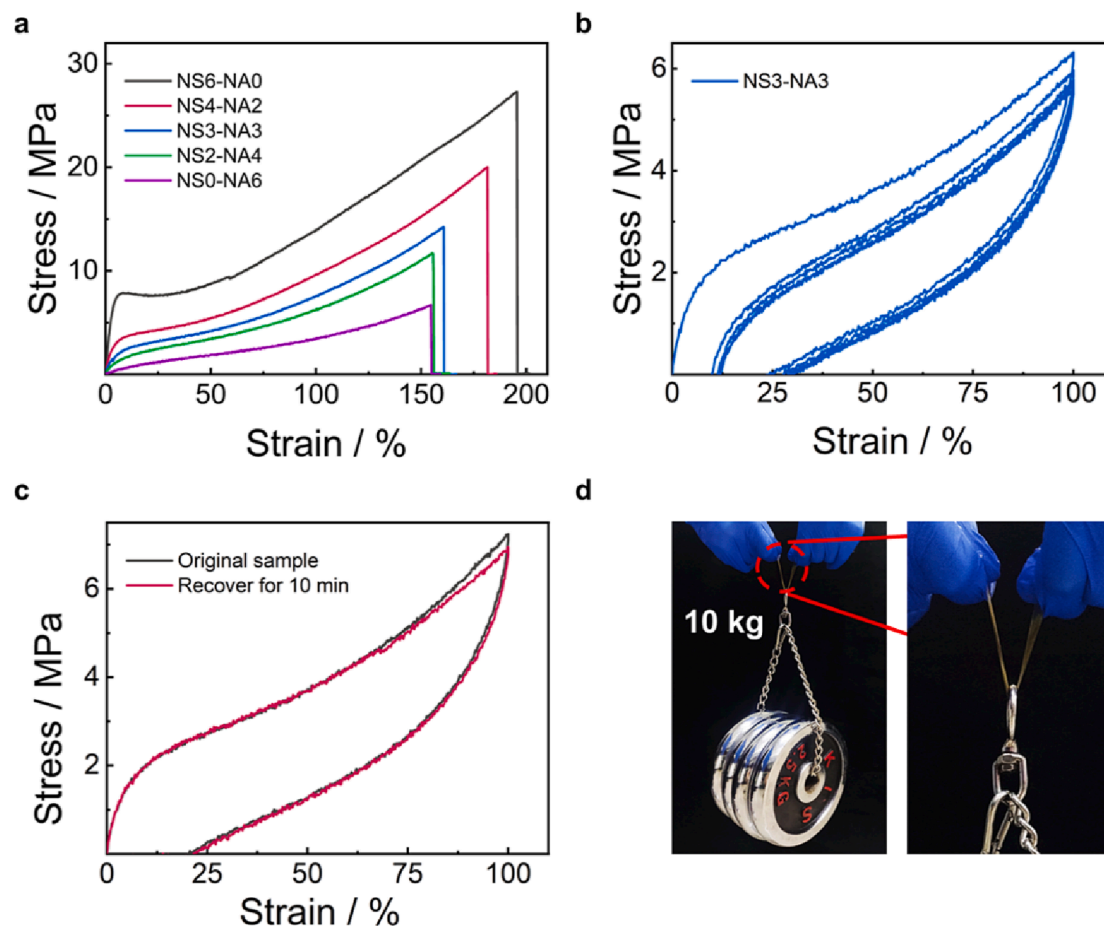


Fig. 1. (a) Stress–strain curves of NS_x–NA_y. (b) Cyclic stretch curve of NS3–NA3. (c) Cyclic stretch curves of NS3–NA3 at the initial state and after a ten-minute recovery. (d) Demonstration of the load bearing of NS3–NA3.

$\tan \delta$ of NS3–NA3 is more than 0.3 in the temperature range of 25–30 °C, indicating that NS3–NA3 has good impact resistance. We also calculated the cross-linking density (ρ) by the classical equation based on the statistical theory of rubber elasticity as shown in Eq. (1): [45]

$$\rho = \frac{E}{3RT} \quad (1)$$

where E is the storage modulus of elastomers in the rubbery plateau region at the temperature T ($T_g + 110$ °C); R is the gas constant. Based on the values of storage modulus (Fig. S7), the corresponding ρ value of NS_x–NA_y is summarized in Table 2. The higher the NS content, the higher the cross-linking density.

To further investigate the dynamic mechanical properties, a dynamic tensile test was conducted on NS3–NA3. Fig. 2c shows the stress–strain curves of NS3–NA3 at different tensile rates. As the tensile rate increases from 2 to 500 mm/min, the tensile strength increases from 10.8 to 16.6 MPa. At the same stretching ratio, the strength at a high stretching rate is higher than that at a low stretching rate. Fig. 2d shows the single-cycle tensile curves of NS3–NA3 at different tensile rates. The higher the strain rate, the larger the area enclosed by the hysteresis curve. In other words, more energy is consumed at a higher strain rate, which is beneficial for impact resistance. This is understandable that the elastomer contains abundant hydrogen bonds and nanoclusters, more energy is required to break through their restrictions when the elastomer encounters an external force. Clearly, the more energy is dissipated when the external force is exerted more quickly.

In addition, a self-constructed falling ball impact device (Fig. 2e) was used to examine the buffer energy absorption capacity of NS_x–NA_y and the maximum impact force was recorded. As shown in Fig. 2f, the

maximum impact force of control sample is 97.5 ± 4.9 N, and that of NS6–NA0 is 52.8 ± 3.5 N, indicating that about half of the impact force is dissipated by the elastomer. As NA content increases, the maximum impact force decreases from 52.8 ± 3.5 N to 33.5 ± 0.37 N, indicating that the impact force is further attenuated. This is understandable because the cross-linking density decreases making it easier for polymer chains to move. When the falling ball contacts with the elastomer, part of the energy is dissipated by the polymer chains motion, thus leading to a lower impact force. Actually, besides the mechanical properties, oligosiloxane nanoclusters can endow NS_x–NA_y elastomers with excellent thermal stability. As shown in Fig. S7, all curves exhibit two stages, which indicates that the samples decomposed mainly in two steps. The first step could be attributed to the degradation of hard segments and telomers. Meanwhile, T_{50s} of NS6–NA0, NS4–NA2, NS3–NA3, NS2–NA4 and NS0–NA6 are 421.2, 407.6, 397.5, 391.7, and 383.7 °C, respectively, higher than that of neat polyurea (369.2 °C). This is because nanoclusters can introduce more rigid benzene ring and the high cross-linking density inhibits the motion of polymer chains [35,46]. On the other hand, the higher the NS content, the higher the thermal residue. This is understandable that higher content of NS endows the material with higher inorganic components.

Fig. 3a shows the XPS spectra of NS_x–NA_y. The peaks at 150 and 100 eV are attributed to Si (2 s and 2 p) in nanoclusters, and the peak at 685 eV belongs to F (1 s) in the fluorocarbon segment of amphiphilic telomers. For NS6–NA0, no F (1 s) peak can be detected, because it does not contain NA. The peak of F (1 s) becomes obvious, even with a NA content of 20 wt%. Table S1 shows that the F content at the surface is significantly higher than that at the bulk of polyurea containing

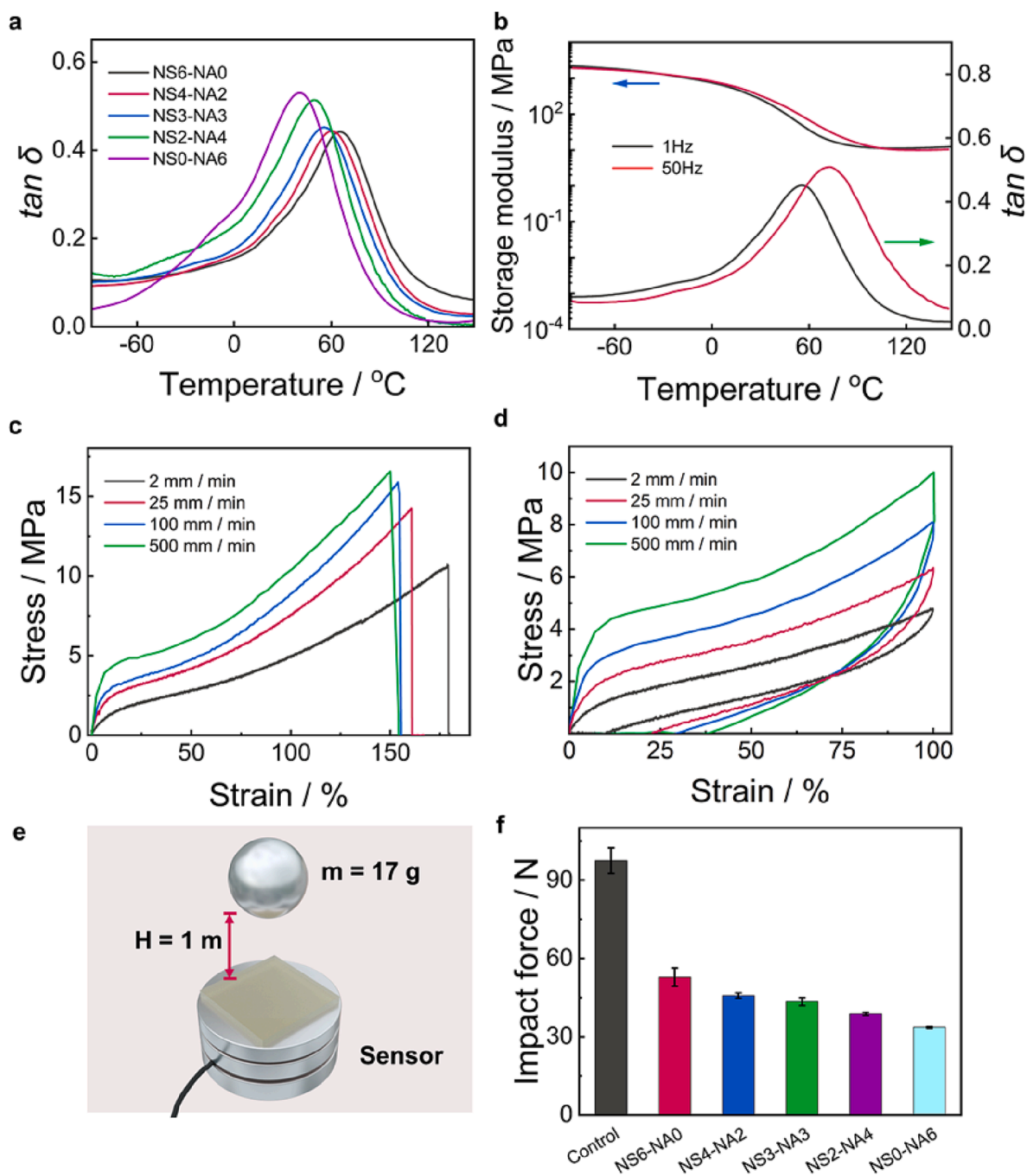


Fig. 2. (a) Dynamic mechanical analysis of NS x -NA y at 1 Hz. (b) Dynamic mechanical analysis of NS3-NA3 at 1 Hz and 50 Hz. (c) Stress-strain curves of NS3-NA3 at different rates. (d) Single-cycle curves of NS3-NA3 at different rates. (e) Schematic diagram of self-constructed falling ball impact device. (f) The maximum impact force of NS x -NA y .

Table 2
Cross-linking density of NS x -NA y .

	NS6-NA0	NS4-NA2	NS3-NA3	NS2-NA4	NS0-NA6
ρ (mol/m ³)	1264	1181	1154	773	580

telomers (NS4-NA2, NS3-NA3, NS2-NA4, and NS0-NA6). As mentioned above, the telomers in NA contain randomly distributed PEG segments and fluorocarbon segments. Attributed to the low surface energy of fluorocarbon segments, the telomers can self-enrich onto the surface [29,30]. In this case, PEG segments have reached the surface with fluorocarbon segments during the curing process, which was

slightly affected by the high cross-linking density. Fig. 3b shows the static WCA of NS x -NA y . Before water immersion, the initial WCA of NS6-NA0 is approximately 90° indicating that the surface is hydrophobic. When NA was added, the WCA of each polyurea elastomer (NS4-NA2, NS3-NA3, NS2-NA4, or NS0-NA6) increased to approximately 105°, indicating that the fluorocarbon segments in NA can migrate to the surface, and the surfaces become more hydrophobic. After soaking in deionized water for 96 h, the WCAs of samples except NS6-NA0 significantly decrease. This fact clearly proved that PEG segments appear on the surface and make the surface more hydrophilic. The enrichment of PEG would endow the surface antifouling function [47].

The protein resistance of NS x -NA y was measured by QCM-D.

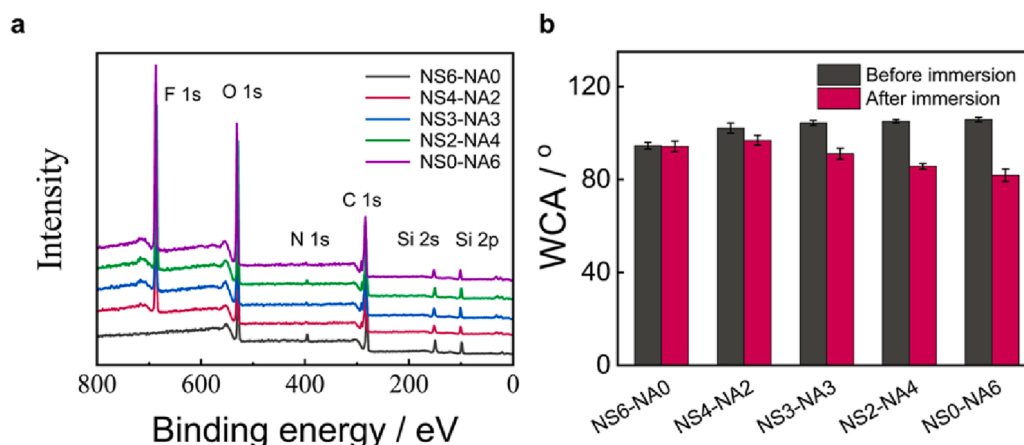


Fig. 3. (a) XPS spectra of NSx-NAy. (b) WCA of NSx-NAy.

Fibrinogen ($M_w = 340$ kDa) is a hydrophobic protein and typically used as a model [48]. Fig. 4a depicts the time dependence of frequency change (Δf) and dissipation change (ΔD) regarding the fibrinogen adsorption on the NSx-NAy surfaces. For NS6-NA0, the remarkable decrease in Δf and increase in ΔD after rinsing with the fibrinogen solution indicate the considerable protein adsorption because it is hydrophobic. For NS4-NA2, variations in Δf and ΔD decrease significantly, indicating the adsorption of protein declines. As NA content increases, the protein resistance increases, where small difference of Δf and ΔD for NS3-NA3, NS2-NA4, and NS0-NA6. As discussed above, PEG in amphiphilic telomers can appear onto the surface forming a hydration layer to effectively resist the adsorption of protein.

To evaluate the antibacterial properties of NSx-NAy, antibacterial assays were performed by using fluorescent dyeing method. Fig. 4b-4d show the results of antibacterial assays and relative bacterial adhesion (RBA) of NSx-NAy. The RBA of *Pseudomonas* sp. can be found in Fig. S5. As depicted in Fig. 4b, many bacteria can be observed on the surfaces of control sample and NS6-NA0. As NA content increases, the number of bacteria on NS4-NA2 drastically decreases, and almost no bacteria (RBA < 10%) exist on the surfaces of NS3-NA3, NS2-NA4, and

NS0-NA6. This is because the fluorocarbon segments with low surface energy can serve as a driving force to draw the telomers to the surface during curing process. Besides, the low surface energy fluorocarbon segments are also capable of easily removing adhered organisms from the surface [31]. When soaked in water, the PEG segments can appear on the surface and form a hydration layer with water through hydrogen bonding, which serves as a barrier and can inhibit the adhesion of bacteria [49,50]. Namely, the anti-biofouling ability is improved as NA content increases. To further evaluate the antifouling performance of NSx-NAy, diatoms, another model fouling organism was also used. The optical images and quantitative density of diatoms adhered on NSx-NAy are shown in Fig S6 and S7. A great density of diatoms can be observed on the surfaces of control sample (210 cell/mm²) and NS6-NA0 (176 cell/mm²). As NA content increases, the diatom density significantly decreases from 210 cell/mm² to 11 cell/mm², indicating that the NSx-NAy elastomers have excellent anti-diatom ability. This result is consistent with those from anti-protein and anti-bacterial assays, indicating that the NSx-NAy elastomers with NA content of 30 wt% or higher have good fouling resistance.

Due to the suitable operation time and viscosity, the NSx-NAy

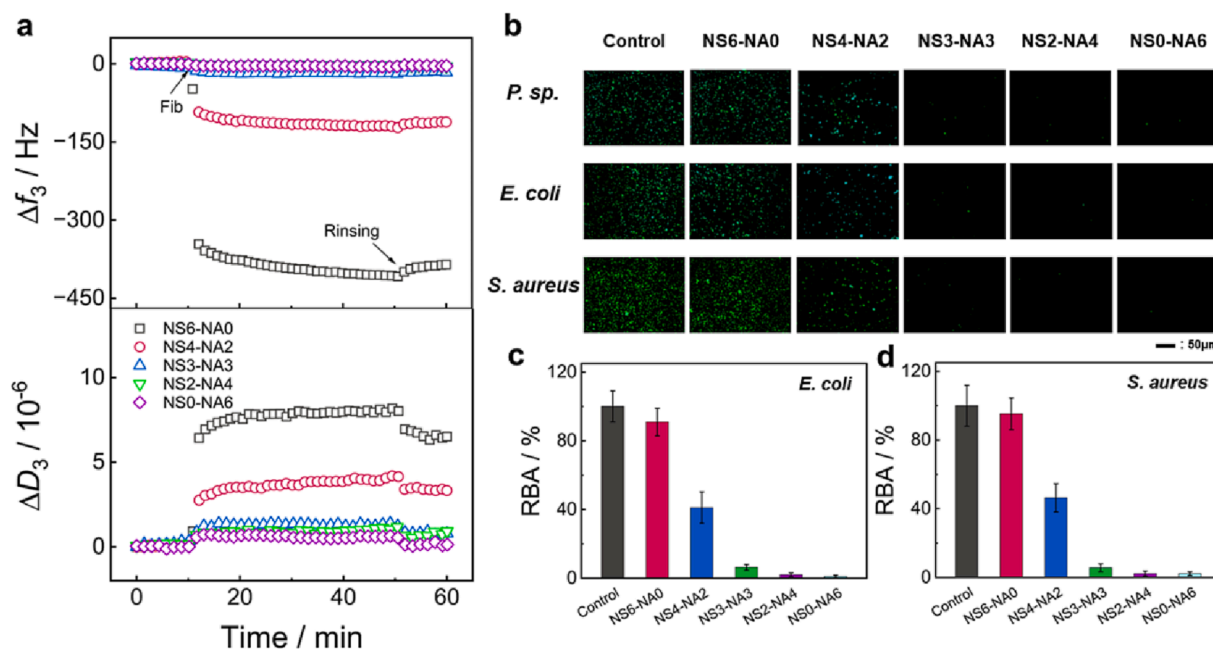


Fig. 4. (a) The QCM-D curves of NSx-NAy. (b) Fluorescence microscopy images of *Pseudomonas* sp., *E. coli* and *S. aureus* on the surfaces of NSx-NAy. (c) RBA of *E. coli*. (d) RBA of *S. aureus*.

solution could penetrate and integrate with the fabric and reinforce it (Fig. 5a). Kevlar fabric with excellent mechanical properties, thermal stability, and flame retardancy is extensively used in the protection field. However, Kevlar fibers tend to delaminate from each other, which makes Kevlar fabric easy to be punctured. As shown in Fig. S12, when hanging 2-kg kettlebell, the polyurea elastomer does not peel off from the surface of the Kevlar fabric. The fact indicates that there is a good interface compatibility between polyurea elastomer and Kevlar fabric due to strong hydrogen bond interaction between the urea group in polyurea and the amide group in Kevlar fabric [51,52]. Coating Kevlar fabric with NSx-NAy is an effective method to improve the puncture resistance. NS3-NA3 was chosen as a typical sample to be combined with Kevlar fabric. Fig. 5b shows the anti-puncture performance of Kevlar and Kevlar/NS1-NA1. Sharp needle was dropped from the same height (50 cm). The one-layer Kevlar fabric sample was easily punctured, while the one-layer Kevlar/NS3-NA3 composite was not punctured. As shown in Fig. 5c, one-, two-, and three-layer Kevlar fabric samples were readily punctured. Nevertheless, one-layer Kevlar/NS3-NA3 composite was slightly deformed without breakage under the same conditions. Instron5966 with a 10-kN load was used to further investigate the puncture resistance of the Kevlar/NS3-NA3 composite (Fig. 5d). One-, two- and three-layer Kevlar fabrics can withstand force of 10.5, 17.4, and 43.3 N at 4.8-, 6.2-, and 9.1-mm needle displacements. In contrast, one-, two-, and three-layer Kevlar/NS3-NA3 can withstand force of 108.8, 207.8 and 563.4 N at 6.3-, 10.0- and 13.3-mm needle displacements, respectively. Fig. 5e shows the puncture resistance force of different layers of Kevlar fabric and Kevlar/NS3-NA3 composite at different puncture depths (2, 4, 6, and 8 mm). At the same puncture depth, the puncture resistance force of the composite is higher than that of Kevlar fabric. In addition, the three-layer Kevlar fabric can only withstand a puncture depth of approximately 9 mm, whereas the three-layer Kevlar/NS3-NA3 composite can withstand a puncture depth of

more than 12 mm, indicating that NS3-NA3 can enhance not only puncture resistance force of Kevlar fabric but also its puncture displacement.

Fig. 6a and 6c show the SEM images of *E. coli* and *S. aureus* on Kevlar fabric (control sample) and Kevlar/NSx-NAy composite. Many bacteria can be observed on the control sample and Kevlar/NS6-NA0 composite. With the introduction of NA, the number of adhesion bacteria gradually decreased, and almost no bacteria can be observed on Kevlar/NS3-NA3, Kevlar/NS2-NA4 and Kevlar/NS0-NA6. As NA content increases, the RBA of two types of bacteria lower to 20% (Fig. 6b and 6d), indicating that the NSx-NAy elastomers can endow Kevlar fabric with antibacterial ability. The fact indicates the NSx-NAy elastomers exhibit good versatility.

4. Conclusion

We have prepared a polyurea elastomer with impact resistance and antifouling performance by using secondary amine-oligosiloxane nanoclusters and a diisocyanate-terminated PTMG-based prepolymer. Attributed to the unique secondary amine-oligosiloxane nanoclusters, the polyurea elastomer has both suitable operation time and excellent mechanical properties. The polyurea elastomer has excellent energy dissipation and impact-resistant capacity due to the synergistic effect of the nanoclusters and hydrogen bonds. The amphiphilic telomer segments grafted on the nanoclusters can migrate to the surface and endow the polyurea elastomer with outstanding antifouling ability. In addition, it exhibits good versatility and can form a composite with Kevlar fabric through a simple dip-coating process. The composite based on Kevlar fiber and polyurea elastomer has good anti-puncture and antifouling capability. This work provides a simple but valuable strategy to prepare high-performance polyurea elastomer. The high-performance polyurea elastomer may find applications in the fields of flexible wear protection,

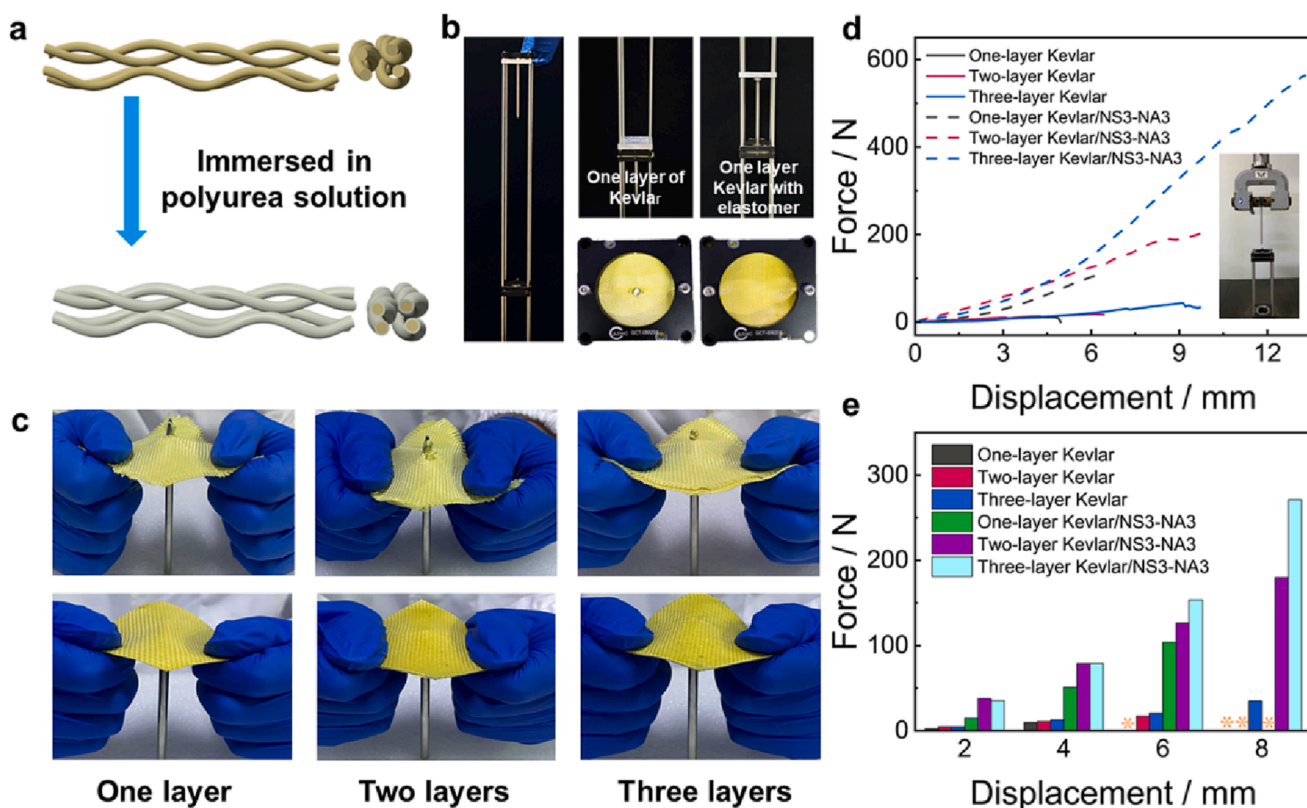


Fig. 5. (a) Schematic of Kevlar/polyurea composite. (b) (c) Anti-puncture displays of the Kevlar fabric and Kevlar/NS3-NA3 composite. (d) Force-puncture displacement curves of the Kevlar fabric and Kevlar/NS3-NA3 composite (one layer, two layers, and three layers). (e) Statistics of puncture force of the Kevlar fabric and Kevlar/NS3-NA3 composite (one layer, two layers, and three layers) at a fixed depth (2, 4, 6, and 8 mm). * It means that the sample has been punctured.

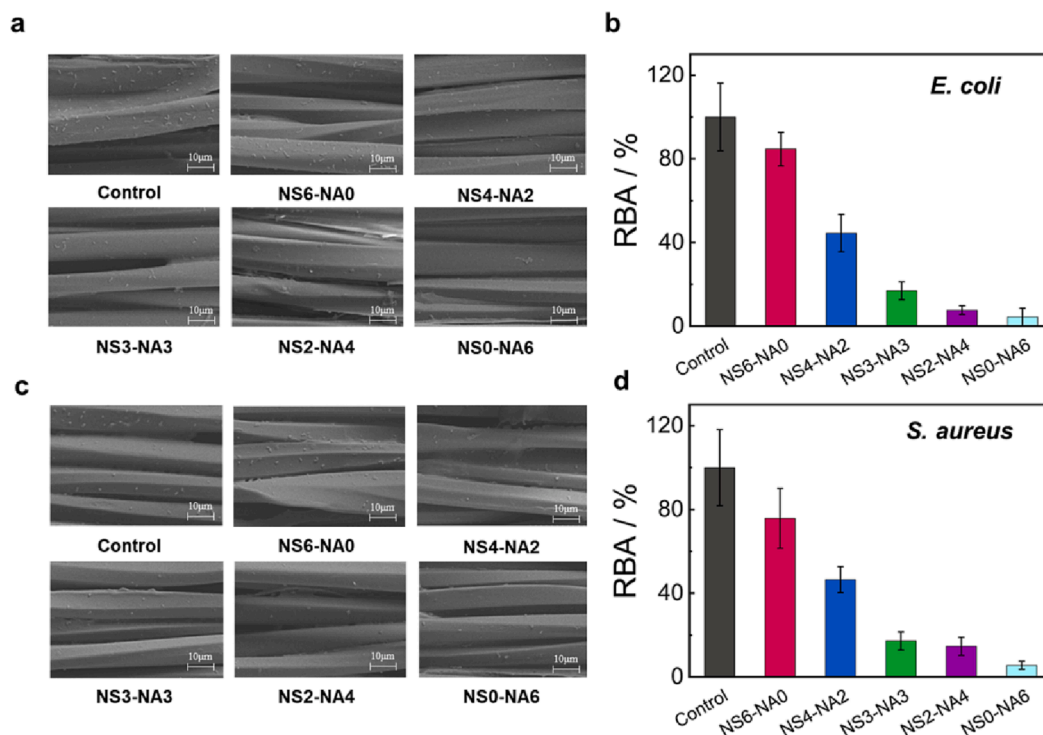


Fig. 6. (a) SEM images of *E. coli* on the Kevlar fabric and Kevlar/NSx–NAy composite. (b) RAB of *E. coli* (c) SEM images of *S. aureus* on Kevlar fabric and Kevlar/NSx–NAy composite. (d) RAB of *S. aureus*.

marine antifouling, and biomedical engineering.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.cej.2023.142769>.

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